Luna Strah 5.14

3 $y:B \vdash const_y : A \to B$

$$\dfrac{\dfrac{}{y:B, x:A \vdash y:B}\,Ax}{y:B \vdash \lambda x:A.\, y : A \to B}$$

rewrite (przepisowalna prawda?)

$const_y\,(f\,z) \longrightarrow y$

$$\boxed{(\lambda_x : A\ t)\,s \longrightarrow t[x \mapsto s]}$$

$const_y\,(f : z) \longrightarrow (\lambda x:A.\, y)(f\,z) \longrightarrow$

$\longrightarrow y[x/f\,z] = y$

$f\,(cost_y\ x) \longrightarrow f\,(y[x \to x]) \longrightarrow f\,y$

$$\vdash \quad A+B \longrightarrow B+A$$

$$\dfrac{\dfrac{}{t:A+B \vdash t:A+B}\,Ax \qquad \dfrac{\dfrac{}{A \vdash A}\,Ax}{A \vdash B+A} \qquad \dfrac{\dfrac{}{B \vdash B}\,Ax}{B \vdash B+A}}{A+B \vdash \quad B+A}$$

$$\dfrac{A+B \vdash \quad B+A}{\vdash \quad A+B \longrightarrow B+A}\,\rightarrow I$$

$$\dfrac{\dfrac{}{t:A+B \vdash t:A+B}\,Ax \qquad \dfrac{\dfrac{}{t:A, B\ a:A \vdash A}\,Ax}{t:A, B\ a:A \vdash in_2(a):B+A}\,+I_2 \qquad \dfrac{\dfrac{}{t:A, B\ b:B \vdash B}\,Ax}{t:A, B\ b:B \vdash in_1(b):B+A}\,+I_1}{t:A+B \vdash \quad \begin{array}{l} \text{case } t \\ \quad |in_1(a)| \longrightarrow in_2(a) \\ \quad |in_2(b)| \longrightarrow in_1(b) \end{array}}\,+E$$

$$\dfrac{}{\vdash \quad \lambda t:A+B.\ \begin{array}{l} \text{case } t \\ \quad |in_1(a)| \longrightarrow in_2(a) \\ \quad |in_2(b)| \longrightarrow in_1(b) \end{array} \quad : A+B \longrightarrow B+A}\,\rightarrow I$$

$$\dfrac{\dfrac{\dfrac{}{t : A\times(B+C) \vdash t : A\times(B+C)}\,\times\mathrm{Ax}}{t : A\times(B+C) \vdash \mathrm{snd}(t) : B+C}\,\times E_2 \qquad \dfrac{}{t : A\times(B+C), b : B \vdash b : B}\,\mathrm{Ax} \qquad \dfrac{}{t : A\times(B+C), c : C \vdash c : C}\,\mathrm{Ax}}{t : A\times(B+C) \vdash \begin{array}{l}\mathrm{case}\ \mathrm{snd}(t)\\ \mathrm{in}_1(b) \mapsto \mathrm{fst}(t), b\\ \mathrm{in}_2(c) \mapsto \mathrm{fst}(t), c\end{array} : A\times B + A\times C}\,+E$$

$$\dfrac{}{\vdash \lambda t.\ \begin{array}{l}\mathrm{case}\ \mathrm{snd}(t)\\ \mathrm{in}_1(b) \to \mathrm{fst}(t), b\\ \mathrm{in}_2(c) \mapsto \mathrm{fst}(t), c\end{array} : A\times(B+C) \to A\times B + A\times C}\,\to I$$

$$\frac{\boxed{\begin{array}{c} x:A \\ \vdots \\ t:B_x \end{array}}}{\lambda x.A.\, t : \prod_{x:A} B_x}\ \Pi I$$

$$\bot := \prod_{P:Prop} P$$

$$\frac{t:\bot}{\bot\text{-elim}(t):P}\ \bot\ elim$$

$$\varphi \vee \psi := \Pi P:Prop.\ (\varphi \to P) \to (\psi \to P) \to P$$

$$\frac{t:\varphi\vee\psi \qquad \boxed{\begin{array}{c} x:\varphi \\ \vdots \\ s:P \end{array}} \quad \boxed{\begin{array}{c} y:\psi \\ \vdots \\ u:P \end{array}}}{\begin{array}{l} \text{match}\cdot t \\ \quad in_1 x \mapsto s \\ \quad in_2 y \mapsto u \end{array} \quad :P}$$

$$\exists x:A.\ \varphi x := \Pi P:Prop\ (\Pi x:A.\ \varphi x \to P) \to P$$

$$\forall x:A.\ \varphi x := \Pi x:A.\ \varphi x$$

$$\varphi \wedge \psi := \Pi P:prop\ (\varphi \to \psi \to P) \to P$$

③ let $2 = \{0, 1\}$

$\varphi 0 \vee \varphi 1 = \exists x : 2 . \varphi x$

$$\varphi 0 \vee \varphi 1 := \Pi P : Prop . (\varphi 0 \to P) \to (\varphi 1 \to P) \to P$$

$$\exists x : 2 . \varphi x = \Pi P : Prop . (\Pi x : 2 . \varphi x \to P) \to P =$$

$$= \Pi P : Prop ((\varphi 0 \to P) \times (\varphi 1 \to P)) \to P =$$

$$= \Pi P : Prop (\varphi_0 \to P) \to (\varphi 1 \to P) \to P$$

④ $\forall x{:}A.\, Px \to Q \to \exists x{:}A.\, Px \to Q$: $P : A \to Prop$, $a : prop$

$$(\Pi x{:}A.\, Px \to Q) \to (\Pi \varphi{:}Prop.\, (\Pi x{:}A.\, Px \to \varphi) \to \varphi) \to Q$$

1. $t : \Pi x{:}A.\, Px \to Q$ Assumption
2. $s : \Pi \varphi{:}Prop\, (\Pi x{:}A.\, Px \to \varphi) \to \varphi$ Assumption
3. $Q : Prop$ Assumption
4. $s\, Q : (\Pi x{:}A.\, Px \to Q) \to Q$ ΠE (2) (3)
5. $s\, Q\, t : Q$ $\to E$ (4) (1)
6. $\lambda t.\, \lambda s.\, s\, Q\, t : *$

$$\mathbb{R}_{\mathbb{N}} : \Pi_{P:\mathbb{N}\to \mathrm{Type}} (P\,0 \to \Pi_{n:\mathbb{N}}. P\,n \to P(S\,n)) \to \Pi_{n:\mathbb{N}}. P\,n)$$

$$\mathbb{R}_{\mathbb{N}}\ P\ P_0\ P_S\ 0 = P_0$$

$$\mathbb{R}_{\mathbb{N}}\ P\ P_0\ P_S\ S\,n = P_S\ n\ (\mathbb{R}_{\mathbb{N}}\ P\ P_0\ P_S\ n)$$

$\mathrm{add} : \mathbb{N} \to \mathbb{N} \to \mathbb{N}$

$\mathrm{Add}\ n : \mathbb{N} \to \mathbb{N}$

- $P_0 :\ \mathrm{Add}\ 0\ ;\ P_0 := \mathrm{id}_{\mathbb{N}}$
- $P_S\ n :\ \mathrm{Add}\ n \to \mathrm{Add}\ S\,n$

  $f \mapsto (m \mapsto S(f(m)))$

$\mathrm{add} := \mathbb{R}_{\mathbb{N}}\ \mathrm{Add}\ P_0\ P_S$

Alternativa

$\mathrm{add}\ n : \mathbb{N} \to \mathbb{N}$

$P_k := \mathrm{Add}_k := \mathbb{N}$

$P_0 = n$

$P_S\ k\ m :=$

$\mathrm{add}\ n := \mathbb{R}_{\mathbb{N}}\ \mathrm{Add}\ P_0\ P_S$

define $\exp : \mathbb{N} \to \mathbb{N} \to \mathbb{N}$

fix n:

$\exp n : \mathbb{N} \to \mathbb{N}$

$P_0^n := S\,0$

$P_S^n\ k\ \underset{\underset{n^k}{\parallel}}{m} := \mathrm{mul}\ n\ m$

$\exp n := R\ (\lambda_.\mathbb{N})\ P_0^n\ P_S^n$

① Define $mul : \mathbb{N} \to \mathbb{N} \to \mathbb{N}$

fix $n : \mathbb{N}$ $\quad$ $mul\ n : \mathbb{N} \to \mathbb{N}$

$P_0 := 0$

$P_S\ k\ m \overset{\text{"}n \cdot k\text{"}}{=} add\ m\ n$

$mul\ n : \mathbb{R}_{\mathbb{N}}\ (\lambda_.\mathbb{N})\ P_0\ P_S$

dolosi katerega tipa so n-ji

$m = k \cdot n$ ko že vemo kaj $k \cdot n$ da,

potem rabimo $n(k+1)$

$$\frac{A : Type \qquad x, y : A}{x =_A y : Prop} \qquad \frac{x : A}{refl_A x : x =_A x}$$

$$J_A : \Pi P : \Pi x, y : A . \Pi p : x =_A y . Type .$$
$$(\Pi x : A . P\ x\ x\ (refl_A\ x)) \to \Pi x, y : A\ \Pi p : x =_A y . P\ x\ y\ p$$

$$J_A\ P\ f\ x\ x\ (refl_A x) \doteq f\ x$$

9 Show $\forall x, y : A\ x =_A y \Rightarrow y =_A x$

na tabli

10 show

$$\Pi x,y,z:A.\ x =_A y \times y =_A z \to x =_A z$$

$$\uparrow$$

$$(x,y,z:A) \to x =_A y \to y =_A z \to x =_A z$$

$$\uparrow$$

$$(x,y:A) \to (z:A) \to x =_A y \to y =_A z \to x =_A z$$

$$\uparrow$$

$$(x,y:A) \to x =_A y \to (z:A) \to y =_A z \to x =_A z$$

$$\uparrow$$

$$P\ x\ y\ p\ :\ (z:A) \to y =_A z \to x =_A z$$

$$f : (x:A) \to \underbrace{P\ x\ x\ \mathrm{refl}\ x}_{:\ (z:A) \to x =_A z \to x =_A z}$$

$$f\ x := \lambda_.\ \lambda p.\ p$$

JP f - je dokaz

$$\frac{A : \mathrm{Type}}{\mathrm{nil} : \mathrm{List}\ A} \qquad \frac{x : A \quad xs : \mathrm{List}\ A}{\begin{array}{c}\mathrm{cons}\ x\ xs : \mathrm{List}\ A \\ x :: xs\end{array}}$$

$$\begin{aligned}
R_{\mathrm{List}_A} : \ & \Pi (P : \mathrm{List}\ A \to \mathrm{Type}). \\
& (P\ \mathrm{nil}) \\
& \to (\Pi\, x : A,\ xs : \mathrm{List}\ A.\ P\ \overbrace{xs}^{\mathbb{N}} \to P\ \overbrace{(x :: xs)}^{\mathbb{N}}) \\
& \to \Pi\, xs : \mathrm{List}\ A.\ P\ xs
\end{aligned}$$

5

$$\mathrm{length} : \mathrm{List}\ A \to \mathbb{N}$$

$$P\ \mathrm{nil} := \mathbb{N}$$

$$P\ xs := \mathbb{N}$$

$$f := \lambda x : A.\ \lambda xs : \mathrm{List}\ A.\ \lambda m : P\ xs.\ S\ m$$

$$p_{\mathrm{nil}} = 0$$

$$p_{\mathrm{cons}}\ m = S\ m$$

$$\mathrm{length} = R_{\mathrm{List}A}\ (\lambda_.\ \mathbb{N})\ \ 0\ \ f\ \ p_{\mathrm{cons}}$$

$\text{Length} : \text{List } A \to \mathbb{N}$

$P : \text{List } A \to \text{Type}$

$P\ [\,] = \mathbb{N}$

$f_1\ \lambda x{:}A\ \lambda xs{:}\text{List}A.\ \lambda l{:}\mathbb{N}.\ l+1)$

$R_{\text{List}A}\ _\ 0$

$: \Pi\, xs{:}\text{List}A.\ \mathbb{N}$

$R_{\text{List}A}\ (\lambda_.\mathbb{N})\ 0\ (\lambda_{:}A.\lambda_{:}\text{List}A.\lambda l{:}\mathbb{N}.\ l+1)$

$: \Pi\, xs{:}\text{List}A.\ \mathbb{N}$

cons list $\quad$ List A → List A → List A

R_{ListA} $(\lambda_.\text{ListA})$ ...

na tabli

Vektorji

$$\frac{}{[\,] : \mathrm{Vec}\ A\ 0} \qquad \frac{x : A \quad xs : \mathrm{Vec}\ A\ n \quad n : \mathbb{N}}{x :: xs\ \mathrm{Vec}\ A\ (S\ n)}$$

$$R_{\mathrm{Vec}A} : \Pi P : \Pi n \in \mathbb{N}.\ \mathrm{Vec}\ A\ n \to \mathrm{Type}.$$

$$P\ 0\ [\,]$$

$$\to \left(\Pi n : \mathbb{N}.\ \Pi x : A.\ \Pi xs : \mathrm{Vec}\ A\ n.\ P\ n\ xs \to P\ (S\ n)\ (x :: xs)\right)$$

$$\to \Pi n : \mathbb{N}.\ \Pi xs : \mathrm{Vec}\ A\ n.\ P\ n\ xs$$

To list $\Pi n : \mathbb{N}\ \mathrm{vec}\ A\ n \to \mathrm{List}\ A$

$$R_{\mathrm{vec}A}\ (\lambda_.\ \mathrm{List}A)\ [\,]_{\mathrm{List}A}$$
$$(\lambda_ : \mathbb{N}.\ \lambda x : A.\ \lambda xs : \mathrm{Vec}A\ n.\ \lambda rep : \mathrm{List}A.\ x :: rep)$$